CMU 17-332/732 | 05-332/632
Guest Lecture · November 9, 2026

Adaptive Sports

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Who am I?

- Patrick Carrington (He/Him/His)
- Associate Professor, HCII
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Human-Computer Interaction Institute

CMU ACCESS



How can we design technology and interfaces that better support people with diverse abilities?

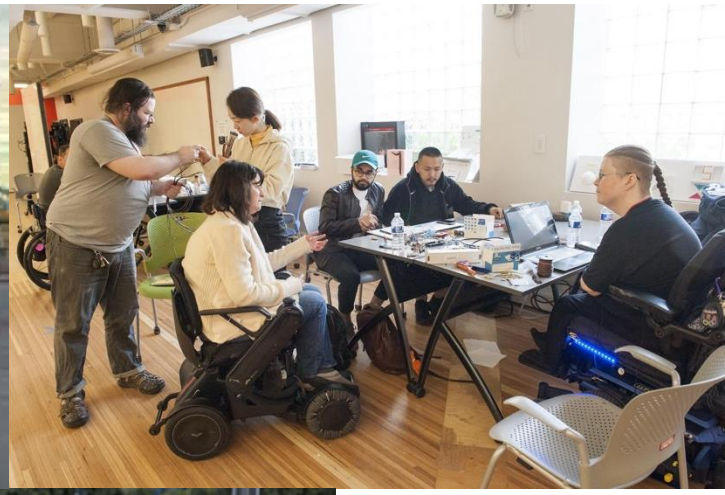
Accessibility + X

What have we done?















Today

- Adaptive sports: what changes and what stays the same
- Endurance sports: running, cycling, kayaking
- Team and winter sports: hockey, skiing, rugby, basketball
- Fairness: classification as a design problem
- Design activity: make your favorite sport more accessible

Accessibility
+
Sports and Recreation

QUESTION



What is your favorite sport?

There are 29 Paralympic sports

- Track and field
- Goalball and blind football
- Swimming
- Powerlifting
- Sitting volleyball
- Para climbing joins at LA 2028 (23 summer + 6 winter)
- Full list: paralympic.org/sports

Running and Cycling





QUESTION



How can you adapt running or cycling?

Think no-tech, low-tech, and high-tech.

Variations and Adaptations - Running





- Blind running with a guide
- Running with prosthetics

Variations and Adaptations - Racing Wheelchair





- Somewhere between running and cycling

Variations and Adaptations - Handcycle





- Pittsburgh hand cyclist Attila Domos (Post-Gazette, 2017)
- Video: Learning About Handcycles

Kayaking



Photo: L.L. Bean



Adaptive
Paddling Gear

creatingability









Soundscape Kayaking Scavenger Hunt!



Microsoft Soundscape:
3D audio cues for wayfinding Kayaking scavenger hunt, October 2018.

Microsoft retired the app in 2023 and released the code as open source.

QUESTION



Consider this case: running blades

Blake Leeper, a double-amputee 400 m sprinter, asked to race against non-disabled athletes

2020: the Court of Arbitration for Sport struck down the rule that disabled athletes must prove no advantage

The same ruling still barred him: his blades made him run several inches taller than allowed

QUESTION



What is fair? What is permissible?

What do you think?



Hockey



Adaptations - Blind Hockey



Blind Hockey Puck vs Standard Hockey Puck

Adaptations - Sled Hockey







Skiing

Adaptive Skiing







TetraSki (ASSETS 2019)





- Powered ski controlled by sip-and-puff, with adjustable shared control
- Alsaleem et al. (2019). Leveraging Shared Control to Empower People with Tetraplegia to Participate in Extreme Sports



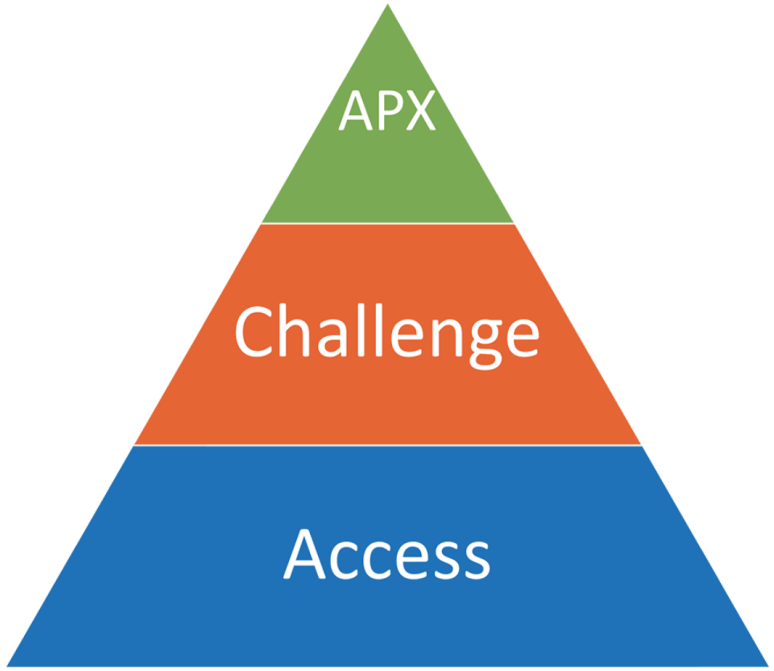
Basketball

Wheelchair Basketball





Classification as a “challenge pattern”



Developed as generative patterns for game design

APX separates access patterns from challenge patterns

Access patterns let more people play

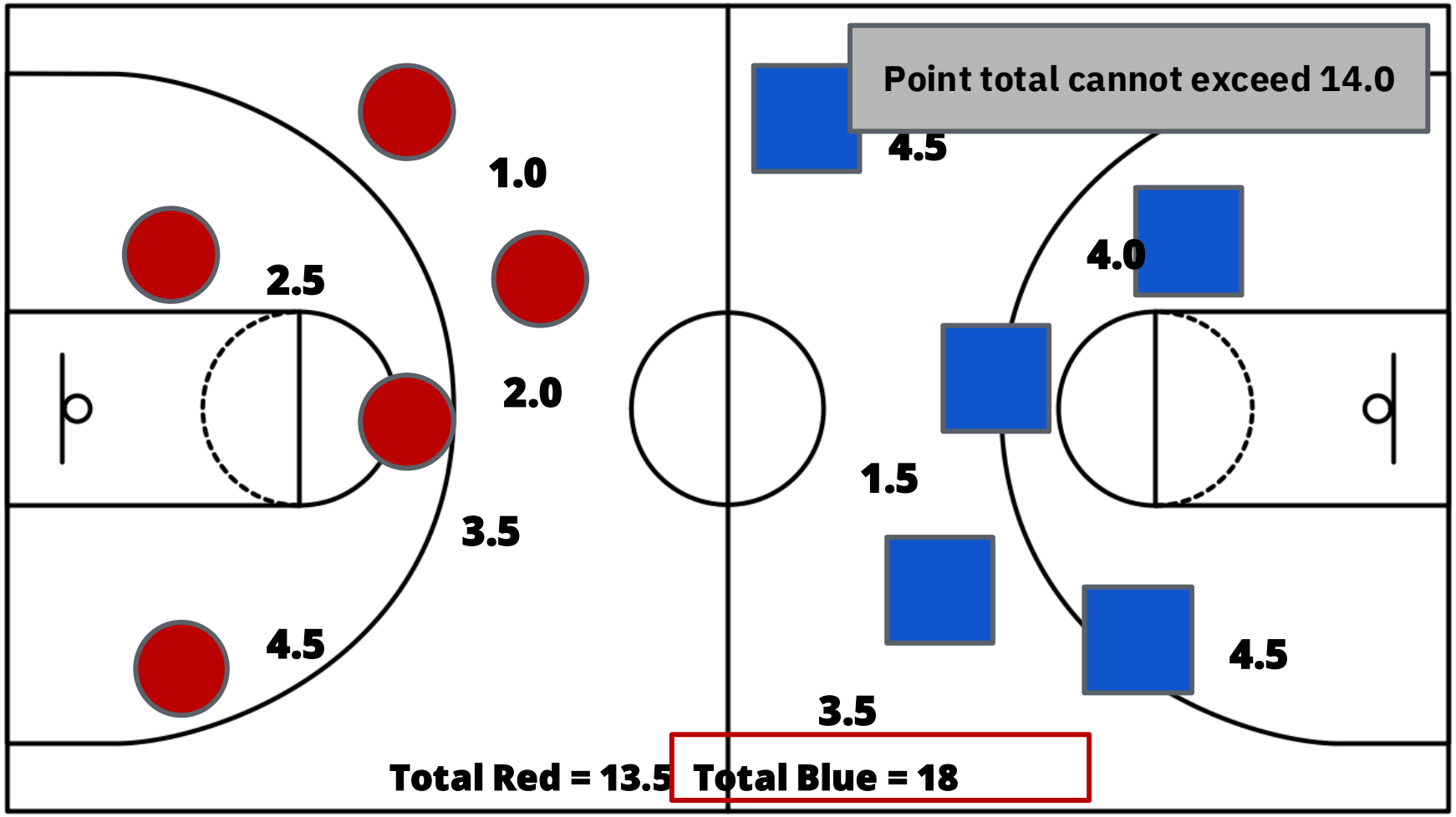
Challenge patterns keep winning meaningful
Classification is sport's challenge pattern

Accessible Player Experiences (APX), accessible.games

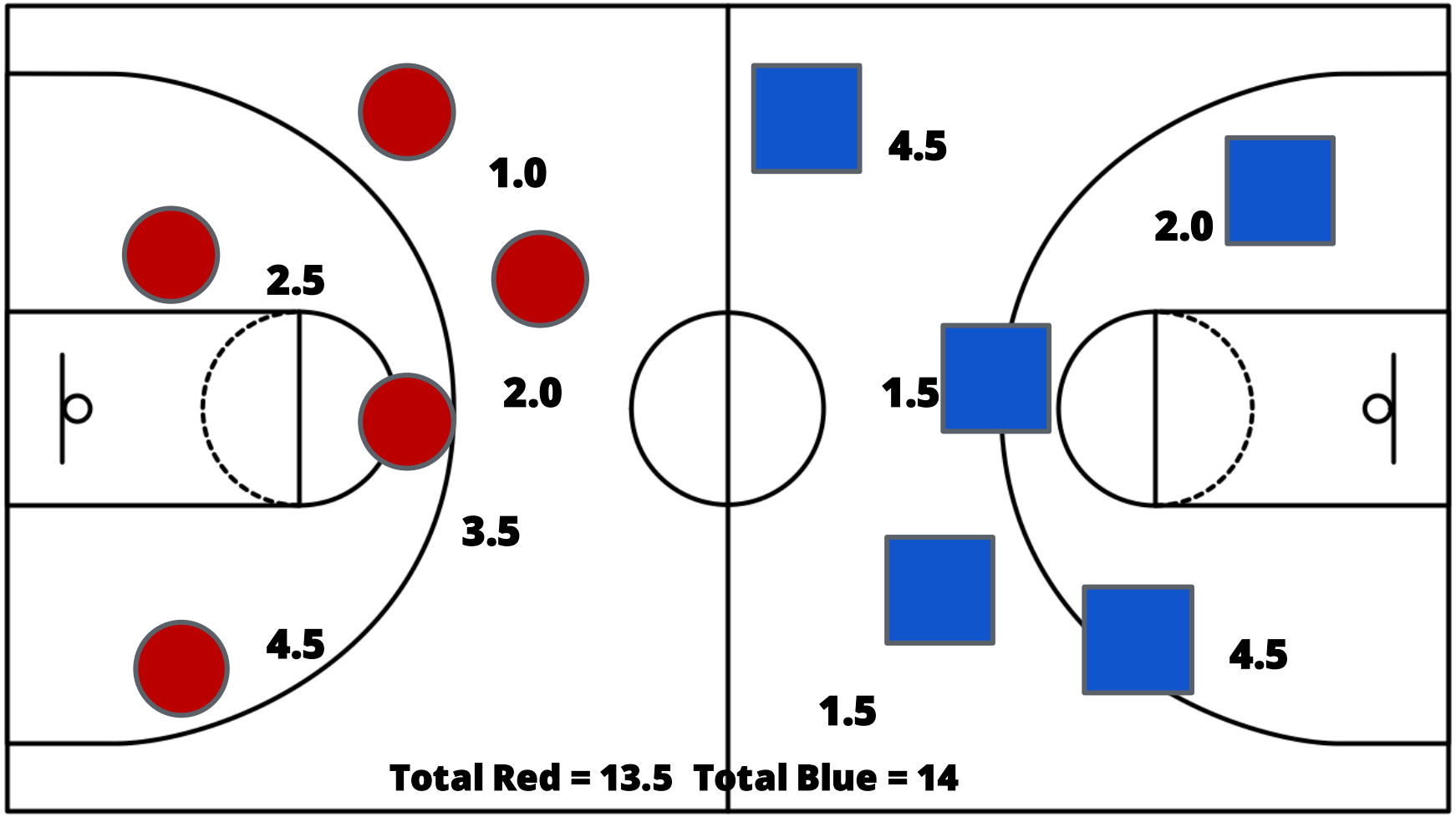
Wheelchair basketball classification

- Every player is classed from 1.0 to 4.5 by trunk function
- 1.0: no active trunk movement forward, backward, or sideways
- 2.5: can lean forward 90° and sit back up without using the arms, but with limited stability
- 4.5: full trunk movement in every plane, no significant weakness
- The five players on court can total at most 14.0 points (IWBF rules, used at the Paralympics)

Is this lineup legal?



A legal lineup



Rugby



Photo: tokyo2020.org

Wheelchair rugby, formerly known as Murderball



Created in 1976, as a more inclusive version of wheelchair basketball

NOT an adaptation of Rugby

Murderball was actually the official name of the sport

Link to Documentary Trailer



How could you make your favorite sport more accessible? More equitable?

In small groups, pick one sport. Name one barrier to access and one fairness question. Sketch a no-tech, low-tech, or high-tech adaptation, then share out.

Today's Accessibility Lunch – 11/9/26 – NSH 3305 @ 12:30 pm

Participation Without Power? A Systematic Review and Call to Center Shared Authority in Accessibility Research
with Joon Jang



Abstract: Participatory methods in accessibility and assistive technology research carry a promise: that disabled people will hold meaningful authority over the technologies designed for them. But to what extent is this promise reflected in how participatory research is actually conducted?...

Free Pizza!

More Readings and sources

- Computing technologies to increase access to physical activity for people with visual impairments. XRDS (2019). dl.acm.org/doi/10.1145/3331067
- Beyond Adaptive Sports. ASSETS ’21. dl.acm.org/doi/10.1145/3441852.3471223
- Alsaleem et al. Leveraging Shared Control to Empower People with Tetraplegia to Participate in Extreme Sports. ASSETS ’19
- Accessible Player Experiences (APX): accessible.games
- International Paralympic Committee sports list: paralympic.org/sports